

Problem set 12: Undirected graphs**Questions**

Q1 Draw the following undirected graph: $G = (V, E, \varphi)$, $V = \{A, B, C, D\}$, $E = \{e_1, e_2, e_3, e_4\}$, $\varphi = \{(e_1, \{A, B\}), (e_2, \{B, C\}), (e_3, \{A, C\}), (e_4, \{C, D\})\}$. Determine $d(A)$, $d(B)$, $d(C)$ and $d(D)$. Draw $\overline{G}$, and draw a different graph isomorphic to $\overline{G}$. Are G and $\overline{G}$ isomorphic?

Q2 Can a 7-vertex simple graph have the following degree sequence?

- (a) 4, 4, 3, 3, 2, 2, 1
- (b) 6, 3, 3, 3, 3, 2, 0
- (c) 5, 5, 5, 2, 2, 2, 1
- (d) 2, 2, 2, 2, 2, 2, 2

Q3 Does a 9-vertex graph exist with the following degree sequence? And simple graph?

- (a) 7, 7, 7, 6, 6, 6, 5, 5, 5
- (b) 6, 6, 5, 4, 4, 3, 2, 2, 1
- (c) 2, 2, 3, 5, 6, 6, 6, 8, 8

Q4 Does an 8-vertex graph exist with the following degree sequence? And simple graph? 6,6,6,6,3,3,2,2

Q5 Prove that any simple finite graph with at least two vertices contains two distinct vertices with the same degree.

Q6 Prove that if a connected graph with at least two vertices has fewer edges than vertices, then it contains a vertex of degree 1.

Q7 Prove that every finite graph in which the degrees of all vertices are at least 2 contains a cycle.

Q8 Let $G = (V, E)$ be a connected graph, and $e \in E$ a bridge. Prove that e cannot belong to any cycle in G .

Q9 Let $G = (V, E)$ be a simple graph with $|V| = 6$. Prove that either G or $\overline{G}$ contains a subgraph that is a 3-vertex complete graph.